Ratul Das

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Personal Profile

Mechanical Engineering Graduate with hands-on experience in CAD modeling, numerical simulation, and Python-based automation. I've experimented with developing simple Python desktop interfaces and have growing interest in GUI-based engineering software. Strong foundation in applied math, numerical methods, and geometric reasoning, supported by hands-on project work using machining and workshop tools. Comfortable adapting to new technical areas and combining computational skills with practical engineering work.

Education

B.Sc. (Engineering) in Mechanical Engineering *(Passing year 2024)*
Shahjalal University of Science and Technology(SUST), Sylhet, Bangladesh

- CGPA: 3.50/4.00
- **Relevant Courses:** Engineering Drawing, Engineering Mechanics, Mechanics of Machinery, Instrumentation and Measurement, Linear Algebra, Applied Engineering Mathematics, and Numerical Analysis.

Skills

Programming Languages: Python, C++, Julia, MATLAB, Octave.

Packages: NumPy, SciPy, Matplotlib, Eigen (C++), PySide (basic).

Software: Ansys Fluent, Ansys ICEM, OpenFOAM, ParaView, Tecplot.

3D Modeling: FreeCAD, SolidWorks, AutoCAD.

Microcontroller: Raspberry Pi, Arduino.

Hardware: Drill Press, Arc Welding, Cut-off saw, Lathe Machine.

Others: LaTeX, Word, Excel, PowerPoint, Mathematica.

Relevant Experience

FreeCAD, Python Scripting for Design Automation and GUI

- Automated FreeCAD workflows using PySide for macro-based geometry generation.
- Designed parametric fluid domains for cyclone and three-phase separators to support iterative CFD studies.
- Exported simulation-ready geometries and integrated parameter sweeps for numerical experimentation.
- Currently developing a GUI-based application using PySide to visualize solutions and the effects of different parameters and numerical schemes for various PDEs. This work extends my Navier–Stokes FDM solver project.

Research Experience

Research Project: Development of an Autonomous Robot for Floating Solid Waste Detection and Collection from Rivers Using Machine Learning Algorithms.

Principal Investigator: Md. Mahmud-Or-Rashid, Assistant Professor, Mechanical Engineering, SUST.
Co-investigator: Md. Syamul Bashar, Lecturer, Mechanical Engineering, SUST.

- Collected and preprocessed image data of water surface waste and designed and built a double-hull catamaran for waste collection.
- Trained, optimized and quantized an SSD MobileNet Lite object detection model for efficient deployment on a Raspberry Pi.

Undergraduate Thesis: Numerical Investigation of the Thermal and Hydraulic Performance of Plain Fin Compact Heat Exchangers Augmented with Constricted Flat Tubes.

Supervisor: Mostafa Rafid, Lecturer, Mechanical Engineering, SUST.

- Validated a numerical model of a heat exchanger by comparing simulation results with published numerical and experimental data.
- Redesigned and optimized tube geometry, resulting in enhanced hydrothermal performance.

Projects

Stepwise Finite Difference Implementation of Convection–Diffusion and Navier–Stokes. link

- Developed individual solvers for 1D and 2D convection and diffusion problems, progressively extending to 2D incompressible Navier–Stokes equations using Python. Applied the solvers to simulate lid-driven cavity and channel flow at low Reynolds numbers, using the projection method, with visualizations to illustrate flow dynamics.

Nonlinear 6-DOF RCAM Flight Dynamics Mathematical Model. link

- Implemented the mathematical equations of a nonlinear 6-DOF rigid-body aircraft model (RCAM) in MATLAB functions and integrated them into Simulink to run time-domain simulations with control inputs and dynamic state outputs.

Derivative Matrix Operator for 2D Laplace Equation via Kronecker Product. link

- Built a sparse matrix-based derivative operator using Kronecker product formulation and incorporated Dirichlet boundary conditions for the finite difference method to solve the 2D Laplace equation.

Mass-Spring-Damper System Simulation using Physics-Informed Neural Network (PINN). link

- Explored how physics-informed neural networks (PINNs) can predict the time evolution of a damped mass-spring system. Studied how different network structures affect training behavior and solution accuracy.

Certifications

Computers, Waves, Simulations: A Practical Introduction to Numerical Methods using Python. link

- Learned finite difference, spectral and element methods for PDEs, implemented mathematical models in Python, and performed benchmarking, convergence tests and visualizations for accuracy.

Deep Learning Specialization. link

- Built and trained deep neural networks, applied TensorFlow for optimization and developed CNNs for image detection, recognition and style transfer.

Industrial Experience

Industrial Trainee
United Energy Limited, Sylhet

Industrial Trainee
Shahjalal Fertilizer Company Ltd. (SFCL), Sylhet

Volunteering

Organising Secretary
Shikorh

In Charge, Engineering Design
Kaizen SUST

Illumination Coordinator
Shikorh

Publications

1. Tripta Sarker, Ratul Das, Sayed Tanvir Ahmed, and Mostafa Rafid. "Numerical investigation of the hydraulic and thermal performance of plain fin compact heat exchangers with modified flat tubes." In *American Institute of Physics Conference Series*, vol. 3307, no. 1, p. 020013. 2025. DOI: 10.1063/5.0262425

2. Sayed Tanvir Ahmed, Ratul Das, Tripta Sarker, and Mahadi Hasan Shanto. "Aerodynamic effects of leading-edge flap angle on NACA 4412 airfoil performance at low Reynolds numbers: A CFD investigation." In *American Institute of Physics Conference Series*, vol. 3307, no.1, p. 020010. 2025. DOI:10.1063/5.0262220

3. Sayed Tanvir Ahmed, Tripta Sarker, and Ratul Das. "Thermohydraulic performance optimization of solar air heater via tailored inverted L-shaped rib: A CFD investigation." In *American Institute of Physics Conference Series*, vol. 3307, no. 1, p. 030005. 2025. DOI: 10.1063/5.0262221